

Readme replication package

Agrivoltaics can reduce political polarization and local opposition to solar energy on land

1 Overview

The code in this replication package cleans and analyzes data from a representative survey among Swiss residents using R. The survey was conducted between 03 May 2024 and 08 June 2024. The file `code/Cleaning.R` loads needed data, cleans it, and recodes variables. The file `code/Analysis.R` loads the cleaned data and produces 30 different tables and 10 plots. The file `code/Conjoint.R` loads the cleaned data and performs a conjoint analysis with 6 tables and 5 plots as output. The replicator should expect the code to run for about 2 hours. This readme file explains necessary steps to run the code.

This reproducibility package contains the following files and folders:

- Folder `data`: Contains all raw data files
- Folder `tables`: The place where all produced tables are saved.
- Folder `plots`: The place where all produced plots are saved.
- Folder `code`: Contains the cleaning script `Cleaning.R`, the script to analyze the data `Analysis.R` and the script `Conjoint.R`. It contains the master script `run_all.R` that runs the entire data cleaning and analysis pipeline.
- `renv.lock`, a file containing information about the environment needed to run the code (more details further down).
- This `Readme.pdf` file
- The codebook `Codebook.pdf` which contains a description of all main variables used in the code.

2 Data

The two main data files `data/data_complete.csv` and `data/data_incomplete.csv` have been obtained through a survey sent out to 16500 residents living in the German and French speaking part of Switzerland between 03 May 2024 and 08 June 2024.

Data on Swiss population statistics were requested from the Federal Statistical Office. They are reflecting the population that the sample used in this paper is drawn from (population living in the German and French speaking part of Switzerland between 18 and 75 years old). They are used to assess the representativeness of our survey data. This data is stored in the following four files:

- data/Age_CH.csv
- data/Gender_CH.csv
- data/Language_ch.csv
- data/urban_rural_ch.csv

3 Software Requirements

- R 4.5.2
 - cregg (0.3.7)
 - tidyverse (2.0.0)
 - ggpubr (0.6.2)
 - kableExtra (1.4.0)
 - stringi (1.8.7)
 - cowplot (1.2.0)
 - FindIt (1.3.0)
 - estimatr (1.0.6)
 - patchwork (1.3.2)
 - texreg (1.39.5)
 - emmeans (2.0.1)
 - scales (1.4.0)
 - vtable (1.4.8)
 - openxlsx (4.2.8.1)
 - dplyr (1.1.4)

When running the code on a local machine in RStudio, dependencies can be managed via **renv** (version 1.1.5). The **renv.lock** file is included in the replication package. After opening the project in R, a working environment can be created by running the following command in the console:

```
renv::restore()
```

If **renv** is not yet installed, first run `install.packages("renv")`.

The code was developed and run on a **10-core Apple M4-based laptop with MacOS version 15.7.3 with 260GB of free space**.

4 Instructions

- Open `code/run_all.R` and adjust the path to the working directory at the top of the script. The working directory should be the parent of the `code` folder.
- Set the working directory by running that line.
- Type `renv::restore()` in the console to initialize a virtual environment containing all the necessary packages listed in the `renv.lock`-file. It might be necessary to enter `1` to initialize the process and to confirm by pressing `y`. If `renv::restore()` does not work, try `renv::init()`.
- Run `code/run_all.R`. This will clean the data and create all relevant output files.
- `code/Cleaning.R`, `code/Analysis.R`, `code/Conjoint.R` can also be run separately.
- To better understand the structure of the code, please press on the “Show document outline”-button in the top right corner of the source code pane in RStudio. Clicking a navigation element jumps to the corresponding code section.